Vibration-Based Structural Damage Detection Using an ARM Cortex-M3 Microcontroller

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***Abstract*— Structural health monitoring (SHM) can detect damage at an early stage. However, most vibration-based SHM systems require expensive data acquisition hardware and computer analysis. The most common damage indicator, natural frequency, is not always effective at detecting faults and will change due to benign factors such as extra mass. Most recent work on TinyML has focused on powerful processing units and there is a lack of fully verified systems using a simple microcontroller. A damage detection system is presented which runs entirely on the LPC1768 ARM Cortex-M3. The MPU6050 MEMS accelerometer samples free vibration of a steel cantilever at 100 Hz. A windowed 1024-point FFT is performed, using parabolic interpolation to obtain peak values, and log envelope damping estimation in the firmware. A decision tree, k-nearest neighbours, 5-8-4 neural network, health score and change detection algorithm are run. All algorithms were trained offline and converted into C code. The neural network achieved 93.8% accuracy (84.2% using a frequency rule) using 89 lab samples. These samples included a healthy state, extra mass, loose joints and loosened supports. The health score had an average error of 5.6 points, and the board and PC agreed on all 89 samples. Damping was used to detect loosened supports rather than frequency. Explainable SHM can be implemented in 20 KB of flash memory on an entry-level microcontroller.**

***Index Terms*—ARM Cortex-M3, edge machine learning, MEMS accelerometer, structural health monitoring, TinyML, vibration-based damage detection**

I. Introduction

Civil and mechanical structures degrade over time due to cracks, corrosion and loose joints. The purpose of structural health monitoring (SHM) is to automatically detect this damage, ideally at an early stage, by identifying if damage exists, locating its position, determining the damage type and severity [1]. The main advantage of vibration-based techniques is that any variation in stiffness, mass or boundary conditions will affect the dynamic response and a single accelerometer can be used [2], [3].

A common metric used is the first natural frequency. Although inexpensive to calculate, there are two main disadvantages. Firstly, many faults cause very little change in this metric and also benign changes, such as added mass or temperature, can have a similar effect to real damage [2]. By using machine learning (ML), multiple features can be combined and the patterns within each condition can be learnt [3], [4], [5]. Most ML-based SHM systems require a PC/cloud system to train and operate, requiring a constant stream of data along with dedicated data acquisition hardware [6], [7].

TinyML takes the inference stage of ML to run on a microcontroller close to the sensor [8]. This has advantages of less data transmission, lower power and cost. However, current literature surrounding edge-based SHM uses multi-core or DSP capable chips and complex neural networks. This project aims to reduce complexity by asking if it is possible to run a fully explainable damage detection pipeline, taking raw acceleration data and producing named faults and a health rating, on an entry-level ARM Cortex-M3 chip without a floating point unit, within 32 KB of code space. The results will also be validated using real hardware.

The contributions of this paper are:

1) An assemble-only cantilever test bench made from two steel rulers, binder clips and a screw clamp, on which three fault types are introduced repeatably and without tools.
2) Fixed-point-friendly firmware for the LPC1768 that samples an MPU6050 at 100 Hz and extracts five features, including interpolated frequency and log-envelope damping, with a self-written radix-2 FFT.
3) Six lightweight models (decision tree, kNN, 5-8-4 neural network, kNN health regression, z-score change detector and novelty detector) trained on a laptop and exported automatically as C code.
4) An experimental study on 89 recordings showing that the ML outperforms a frequency-only rule, that damping is the only usable indicator of support loosening, and that board and PC outputs are identical.

II. Literature Review

A. Vibration-based damage identification

Hou and Xia [2] reviewed damage detection methodologies using vibrational responses from civil engineering structures. The review classifies methodologies according to the vibration features used (natural frequencies, mode shapes, damping and frequency response functions), with frequency being the simplest to measure but providing little information on the nature or position of damage. Gharehbaghi et al. [1] provide a formal definition of the levels within SHM and compare

model-based to data-driven methodologies. Both highlight the issues with environmental and operational variability that can disguise or mimic damage. This highlights why an added mass test was performed for this work.

B. Machine learning for SHM

The transition from manually set thresholds to ML-based solutions using deep learning has been described [3]. A feature extraction stage with a classifier is the most popular form of structure. A comprehensive review of supervised and unsupervised algorithms in civil SHM was carried out [4]. The main issue with implementing these methods is that the available data sets are small. ML provides a novel approach to SHM which can be applied to the identification of damage, detection of anomalies in data and provide an assessment of the condition [5]. A review of ML within SHM was completed alongside emerging sensing technologies [6], such as low-cost MEMS sensors and IoT nodes. A study which is similar to this study was conducted using kNN, support vector machines, decision trees, random forests and naive Bayes algorithms to classify damage in cantilever beams [10]. When only natural frequencies were considered as features the accuracy of the models varied from about 59% to 94%. However, using more complex features from the response the accuracy improved to around 95% using the experimental data.

C. Loosening as a fault

Loose supports and loose bolted joints are among the most frequent problems in real structures. Many studies have been carried out on detecting looseness in threaded fasteners [11]. Vibration-based techniques monitor the change in stiffness and the energy lost at the contact interface, which show up as changes in frequency and damping [11]. This is why damping has been selected as a feature.

D. Frequency estimation

The frequency can only be determined up to $\Delta f = F_s/N$ because the record has a finite length N. A much more accurate estimation can be obtained by fitting a parabola to the three highest samples of the DTFT around the peak of interest. This has minimal additional computational complexity [12] and is applied on the board.

E. SHM at the edge and TinyML

Zonzini *et al.* [7] propose a sensor-to-cloud SHM architecture in which MEMS sensor nodes are used to pre-process data before sending it. Ray [8] and Capogrosso *et al.* [9] review TinyML frameworks, model compression and deployment on microcontrollers. Specific to SHM, Burrello *et al.* [13] train tiny autoencoders to detect anomalies near the sensor, Moallemi *et al.* [14] accelerate output-only system identification on a parallel RISC-V microcontroller, and Capello *et al.* [15] use neural architecture search to fit SHM models on resource-constrained devices.

F. Research gap

Table I shows a summary of the most closely related work. Previous edge-based SHM work has used powerful processors and either anomaly-only or black-box models. The beam classification work has been done using a PC. No previous work has classified fault types, given severity scores and implemented a change detection algorithm using a simple Cortex-M3 processor. This work has been tested using real data with identical outputs from the PC and the board.

TABLE I
COMPARISON WITH CLOSELY RELATED WORK

Work	Platform	Output	Features	Data
[10]	PC	fault class	freq., FRF	sim. + exp.
[13]	MCU (multi-core)	anomaly	raw signal	real
[14]	RISC-V MCU	modal param.	AR/ARMA	real
[15]	MCU	anomaly	learned	real
This work	Cortex-M3 (no FPU)	class, health, change, novelty	freq., damping, 3 more	real (89 flicks)

III. METHODOLOGY

A. Overall approach

The methodology used is shown in Fig. 1. Each labelled flick is stored on the board and then transferred to the laptop via UART. The laptop then uses the same 5 features as the board to train and cross-validate the models. Once trained, they are exported into a C header file (model.h) which is then compiled into the firmware. All new flicks are then classified using only the board. Finally, a parity check runs the same C code on the PC over every recording and compares its output with Python.

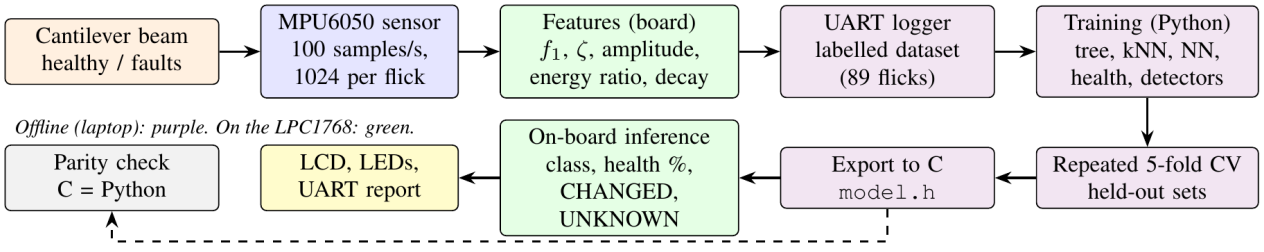


Fig. 1. Method approach: data collection, offline training and on-board deployment.

B. Test structure and fault cases

The structure is a cantilever made out of two 30 cm steel rulers (Fig. 2). Ruler A projects 18.5 cm beyond the table edge and is held by a 10-inch screw G-clamp over a cardboard pad. Ruler B overlaps A by 13.5 cm and is joined to it by three binder clips, giving a free length of about 35.5 cm. The MPU6050 module is taped near the tip. Each measurement starts with a manual flick of the tip, after which the beam rings freely. The build needs no drilling or cutting, so the bench can be reassembled in any lab.

Three fault types were introduced (Fig. 3). *Added mass*: a Rs 20 and a Rs 5 coin taped together at the 28-30 cm mark. This is a change but not damage. *Loose joint*: the middle binder clip removed (level 1), then one end clip removed as well (level 2). *Loose support*: the G-clamp screw loosened in three steps of about 1/4, 1/2 and 3/4 turn; 3/4 turn is the loosest setting at which the beam still stays in place.

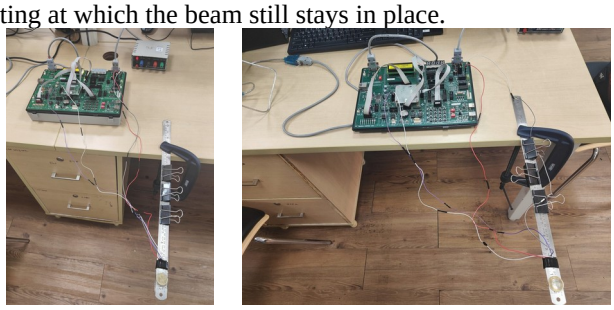


Fig. 2. Experimental setup. Left: LPC1768 board, power supply and the clamped two-ruler cantilever with the MPU6050 near the tip. Right: side view showing the three binder clips, the G-clamp and the wiring to the board.

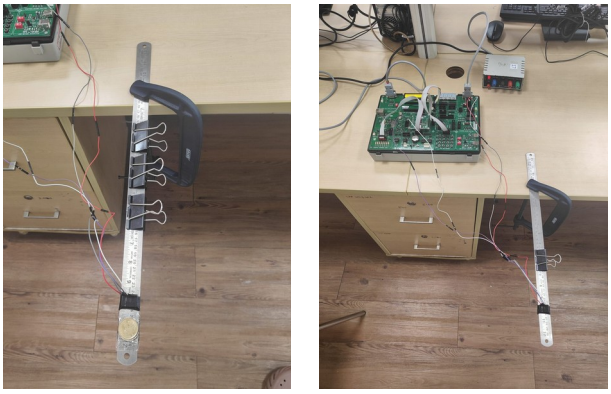


Fig. 3. Fault cases. Left: added mass, coins taped beyond the sensor. Right: loose joint, clips removed from the ruler overlap.

C. Hardware

Fig. 4 shows the block diagram. The LPC1768 (ARM Cortex-M3, 100 MHz, 512 KB flash, 64 KB SRAM) sits on the ALS-SDA-ARMCTXM3-01 development board. The MPU6050 accelerometer is read over a software I²C bus on two GPIO pins, with its measuring range set to ± 4 g, so it can read accelerations of up to four times gravity in either direction, and its internal low-pass filter set to 44 Hz to prevent aliasing. Timer 0 triggers a reading every 10 ms (100 samples per second). The board's 16×2 LCD shows the result. Eight LEDs form a health bar. The 4×4 keypad selects calibration, teach mode and display pages. UART0 at 19200 baud sends every result and, while logging, the raw samples to the laptop.

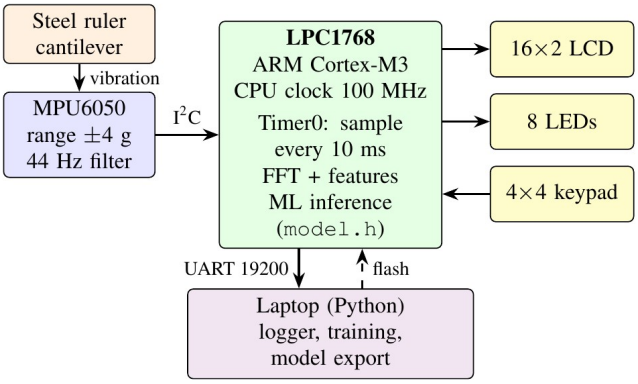


Fig. 4. Block diagram of the hardware.

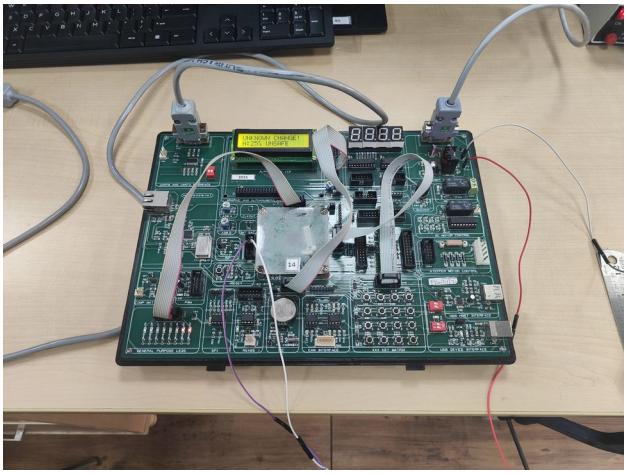


Fig. 5. LPC1768 board during a fault test. The LCD shows the change alarm and the health score.

D. Signal processing and features

A flick is detected when the acceleration moves more than 0.3 g away from its resting value. The next $N = 1024$ samples are then recorded at $F_s = 100$ Hz, which takes 10.24 s. Physically, the method can be understood using single degree of freedom theory for the first mode of the beam,

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}, \tag{1}$$

so a loss of stiffness k, or a gain in mass m lowers the natural frequency [2]. After removing the mean, the record is multiplied by a Hann window to reduce spectral leakage,

$$w[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{N-1}\right), \tag{2}$$

and transformed with a radix-2 FFT,

$$X[k] = \sum_{n=0}^{N-1} x[n]\,w[n]\,e^{-j2\pi kn/N}. \tag{3}$$

The bin width is $\Delta f = F_s/N = 0.098$ Hz, and the largest bin k_p between 3 and 40 Hz is refined with parabolic interpolation through its magnitude β and its neighbours α and γ [12]:

$$\delta = \frac{1}{2}\frac{\alpha-\gamma}{\alpha-2\beta+\gamma}, \quad f_1 = (k_p+\delta)\,\Delta f . \tag{4}$$

Damping is taken from the free decay. For a lightly damped mode the amplitude decays as $A(t) = A_0 e^{-\zeta\omega_1 t}$ with $\omega_1 = 2\pi f_1$. Taking logarithms gives a straight line, so the damping ratio is

$$\zeta = \frac{-s}{2\pi f_1}, \tag{5}$$

where s is the least-squares slope of $\ln$ RMS against time, computed over 0.25 s windows from the loudest window until the RMS falls below 5% of its peak. Three further features are the peak amplitude a_{max} (flick strength), the energy ratio

$$E_r = \frac{\sum_{k\in B}|X[k]|^2 - \sum_{|k-k_p|\le h}|X[k]|^2}{\sum_{k\in B}|X[k]|^2}, \tag{6}$$

which measures energy outside ± 1 Hz of the main peak (rattle and harmonics) within the band B of 3–40 Hz, and the decay time t_d until the RMS drops below 10% of its peak. The feature vector is $x = [f_1, \zeta, a_{max}, E_r, t_d]$.

E. Models

All models use the standardised features $\tilde{x}_i = (x_i - \mu_i)/\sigma_i$.

Frequency-only baseline: a depth-2 tree on f_1, representing the classical frequency-shift rule.

Decision tree: CART of depth 3 with Gini impurity $G = 1 - \sum_c p_c^2$; it is exported as nested `if` statements.

k-nearest neighbours: $k = 3$ with Euclidean distance over the 68 stored training flicks; new examples can be added on the board in teach mode.

Neural network: a 5-8-4 multilayer perceptron,

$$h = max(0, W_1\tilde{x} + b_1), \quad p = softmax(W_2 h + b_2), \tag{7}$$

with 84 parameters; the largest p_c is the predicted class and its value the confidence.

Health score: each training condition has a damage level $d \in [0,1]$ (healthy 0, added mass 0, loose joint 0.5 and 1.0, loose support up to 0.75). The health of a newly done flick is

$$H = 100\left(1 - \frac{1}{3}\sum_{j=1}^{3} d_j\right), \tag{8}$$

where d_j are the levels of its three nearest neighbours. $H < 40$ is reported as UNSAFE.

Change detector: the distance from the healthy baseline,

$$z = \sqrt{\frac{1}{5}\sum_{i=1}^{5}\left(\frac{x_i - \mu_i^h}{\sigma_i^h}\right)^2}, \tag{9}$$

flags CHANGED when $z > 3$. The baseline μ^h, σ^h can be recalibrated on the board from ten healthy flicks (key C). *Novelty detector*: UNKNOWN is shown when the nearest stored flick is farther than twice the 95th percentile of nearest-neighbour distances in the training set.

Trend: the health of successive inspections is fitted with a line $H = a + bt$, and the step at which H reaches 40 is predicted as $(40 - a)/b$.

F. Firmware flow

Fig. 6 shows the firmware loop. All models are compiled from the exported header, so no training happens on the board.

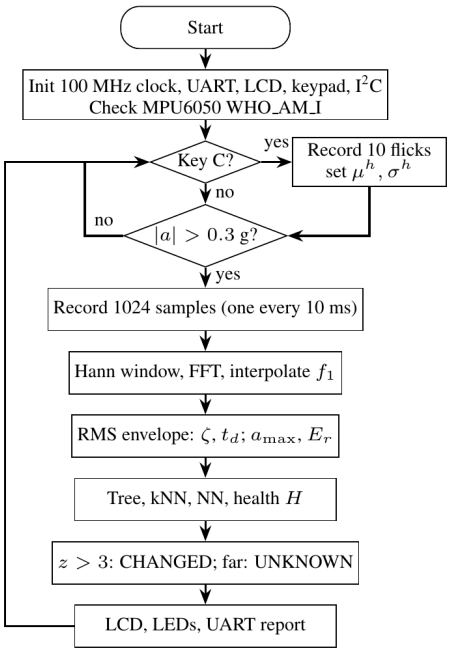


Fig. 6. Flowchart of the on-board firmware.

G. Data collection and evaluation

Each flick is labelled on the laptop and stored with the raw sample data. The data set can be seen in Table II. Prior to each fault, a check was performed on the beam configuration to ensure it was identical to that of a healthy device. Any flicks that resulted in an incorrect overlap of 15 cm between the two rulers were discarded (24 in total). The first two steps corresponding to the loosening of the supports were also excluded from the training data set as the feature values were within the healthy range (Section IV-C). However, they were included in the hold-out test data along with five healthy verification flicks. The accuracy is calculated by using 5-fold stratified cross-validation repeated 5 times on the 68 training flicks.

TABLE II
RECORDED DATASET (MEDIANS)

Condition	Flicks	f_1 (Hz)	ζ (%)	Use
Healthy	30	4.74	2.92	train
Added mass	12	3.68	2.58	train
Loose joint 1	10	4.47	2.67	train
Loose joint 2	8	4.45	1.16	train
Loose support 3/4 turn	8	4.75	3.45	train
Loose support 1/4 turn	8	4.71	3.26	held out
Loose support 1/2 turn	8	4.75	3.13	held out
Healthy re-check	5	4.68	2.95	held out
Total	89			

IV. RESULTS AND DISCUSSION

A. Frequency estimation on the board

Table III compares three frequency estimators against a 16 × zero-padded FFT reference over all 89 flicks, with execution times measured on the LPC1768 using SysTick. Parabolic interpolation reduced the mean error from 0.027 Hz to 0.009 Hz, less than a tenth of the bin width, at no measurable extra time. Zero-crossing counting is 28 times faster but its mean error of 0.34 Hz is larger than the 0.27 Hz shift caused by a loose joint, so it cannot separate the classes (Fig. 7). The FFT takes 117 ms, which is negligible against the 10.24 s recording.

TABLE III
FREQUENCY ESTIMATORS MEASURED ON THE LPC1768

Method	Mean err.	Max err.	Time	RAM
FFT peak bin	0.027 Hz	0.052 Hz	116.9 ms	8 KB
Interpolated FFT, (4)	**0.009 Hz**	0.063 Hz	116.9 ms	8 KB
Zero-crossing	0.338 Hz	2.07 Hz	4.2 ms	none

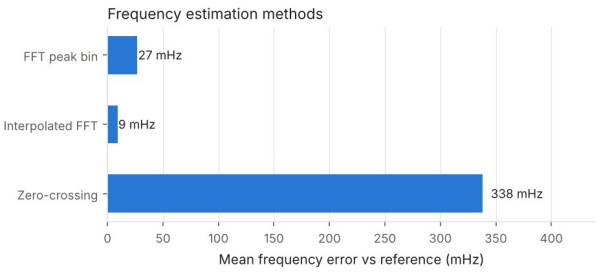


Fig. 7. Error and execution time of the three frequency estimators.

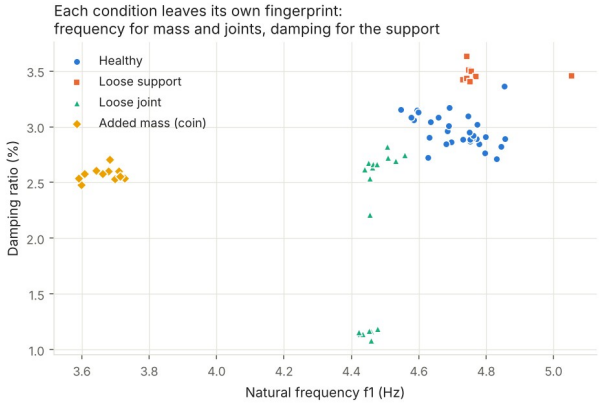


Fig. 8. Damping against first natural frequency for all training flicks.

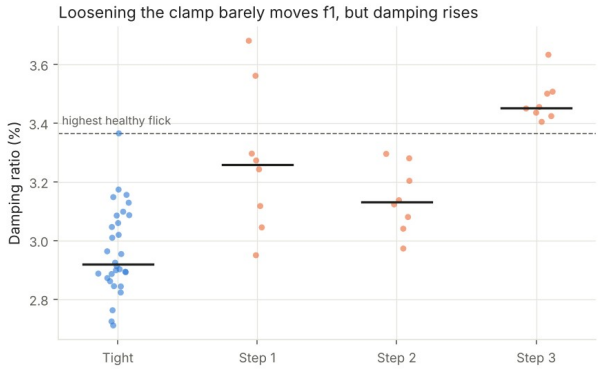


Fig. 9. Damping ratio at each support-loosening step compared with the healthy range.

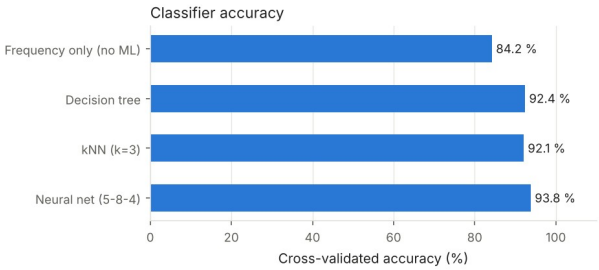


Fig. 10. Accuracy of the frequency-only rule and the three classifiers.

B. Each fault has its own signature

Fig. 8 shows the damping and frequency of each flick. The added mass reduced f_1 by 22%, as predicted by (1). Removing clips decreased f_1 by around 6% due to a loss of stiffness in the joint. Removing the second clip also decreased damping from 2.9% to 1.2%, because less friction remains at the overlap. When the support is loosened there is no change in f_1 (4.75 Hz against 4.74 Hz), but damping increases because the loose end loses energy through friction. This matches the loosening literature [11]. This fault cannot be seen using frequency alone.

C. Detection limit for support loosening

Fig. 9 shows damping at each loosening step. At 1/4 and 1/2 turn the damping stays within the healthy range. For 3/4 turn every flick exceeds the maximum value found in the

normal range by a median of +18%. The screw clamp also maintained a tight grip on the ruler until it was almost released, so the first two steps were below the detection threshold of this system. On these held-out flicks, the neural network labelled 4 of 8 and 3 of 8 as loose support. The change detector triggered once. This is consistent with the features being on the edge of the cluster of normal data.

D. Classification accuracy

Table IV and Fig. 10 give the cross-validated accuracy. All three ML models beat the frequency-only rule by 8 to 10 percentage points. The decision tree uses f_1 for 71% of its splits and damping for 27%, confirming that the gain comes from the damping feature. The confusion matrices in Fig. 11 show that most remaining errors are between healthy and loose support, the hardest pair. All five healthy re-check flicks were classified as healthy with no CHANGED alarm.

TABLE IV
CROSS-VALIDATED CLASSIFICATION ACCURACY (4 CLASSES)

Model	Accuracy
Frequency only (depth-2 tree on f_1)	84.2%
Decision tree, depth 3	92.4%
kNN, $k=3$	92.1%
Neural network 5-8-4	**93.8%**



Fig. 11. Confusion matrices from cross-validation (left to right: frequency only, decision tree, neural network).

E. Health score and trend

The kNN health score had a mean absolute error of 5.6 points, against 17.8 points for linear regression on the same features (Fig. 12). In leave-one-out tests, healthy flicks scored 95.8 on average and none fell below the unsafe threshold. Over the support-loosening steps the mean health was 97, 66, 69 and 33. A line fitted to steps 0 to 2 predicted the beam would become unsafe at step 3.6; in the experiment it was unsafe at step 3, so the forecast was slightly late but in the right range (Fig. 12, right).

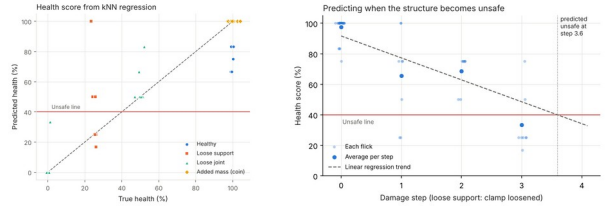


Fig. 12. Left: predicted against true health score. Right: health over successive loosening steps and the predicted unsafe point.

F. On-board operation

The full firmware, including the FFT, all models, and training data for the kNN algorithm requires approximately 20 KB of flash memory to store. This is well within the 32 KB limitation of the free Keil toolchain. The FFT buffers also require 8 KB of RAM. The C code was run through all 89 flicks on the PC and produced the same features and classifications as the Python reference implementation. Fig. 5 shows the board detecting a change and an unsafe health value during a fault test.

G. Contributions verified through results

Each contribution is backed by a measured result. The assemble-only bench allowed repeatable results when checking the healthy flick. The frequency was found to have a difference of 0.06 Hz between the healthy check after all faults had been fixed (4.68 Hz) and the original frequency (4.74 Hz). All 5 healthy checks were found to be healthy. Using the interpolated FFT algorithm on the Cortex-M3 to calculate frequency gave an average error of 0.009 Hz against the reference frequency. This is less than a tenth of the 0.098 Hz bin width and three times better than taking the peak bin alone (0.027 Hz). This was achieved in 117 ms. Adding damping and 3 additional features to the model improved the accuracy of 4 classes from 84.2% (classical frequency method) to 93.8% (neural network). Damping alone showed that when support loosening occurred it did not affect the frequency but did increase damping by 18%. The kNN model gave a good prediction of the severity of damage with an average error of 5.6 points. All healthy flicks had a score above the unsafe level. The trend prediction shows that the unsafe point would occur within one step of the actual unsafe point. Exporting the models to C kept the same features and logic as Python, giving identical results on all 89 flicks, and everything fits in about 20 KB of flash memory. An honest upper limit was found during this experiment. A support loosening of less than about 3/4 turn could not be differentiated from a healthy state using this sensor and structure.

H. Limitations

The dataset is small (89 flicks) and comes from one structure in one room, so temperature and long-term drift were not studied. Flicks were applied by hand, which varies the amplitude; the amplitude feature absorbs this, but a mechanical exciter would give cleaner data. Only the first mode was used.

V. CONCLUSION

This paper demonstrates that a full damage detection system using vibration data, producing a fault name, health value and trend prediction from raw acceleration data can be implemented on an entry-level ARM Cortex-M3 processor and low-cost MEMS accelerometer. The tool-free cantilever test rig gave an accuracy of 93.8% in distinguishing between healthy, added mass, loose joint and loose support cases, compared to 84.2% using the classical frequency shift rule. The health value had an average error of 5.6 points. The main conclusion was that damping, which is not considered by the

frequency rule, was the only usable source of information regarding loose supports. All models fit within approximately 20 KB of flash memory and behave identically to the PC version. In the future the system will be improved by implementing a mechanical exciter, temperature compensation, higher modes from a second sensor and more data (over multiple days) to reduce the detection limit for early loose support.

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